

Sonia Raychaudhuri

Email | [Portfolio](#) | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

PhD Advisor – [Angel X. Chang](#) | Simon Fraser University [3DLG] [GrUVi]

Embodied AI, semantic navigation, multimodal reasoning, semantic mapping, LLM finetuning, LLM evaluation, prompt engineering, RAG



Summary

I am an Embodied AI researcher specializing in robot navigation and multimodal spatial reasoning, with a focus on building robust human-centric autonomy for real-world environments. I received my Ph.D. from [SFU](#) in February 2026, advised by [Angel Chang](#). My work aims at building autonomous real-world systems by bridging vision, language and spatial representations and enabling robots to understand instructions, navigate efficiently and generalize to unknown environments.

Skills

- Embodied AI • Semantic Navigation • Instruction Following • Multimodal Reasoning
- Natural Language Understanding • Reinforcement Learning • Deep Learning • Python • PyTorch

Research Highlights

- 1 S. Raychaudhuri, D. Ta, K. Ashton, A. X. Chang, J. Wang, and B. Bucher, *Zero-shot Object-Centric Instruction Following: Integrating Foundation Models with Traditional Navigation*, 2025. arXiv: [2411.07848](#). URL: <https://sonia-raychaudhuri.github.io/nlslam/>.
- 2 S. Raychaudhuri, S. Wani, S. Patel, U. Jain, and A. Chang, “Language-Aligned Waypoint (LAW) Supervision for Vision-and-Language Navigation in Continuous Environments,” in *EMNLP*, 2021. URL: <https://3dlg-hcvc.github.io/LAW-VLNCE>.
- 3 S. Raychaudhuri, T. Campari, U. Jain, M. Savva, and A. X. Chang, “MOPA: Modular Object Navigation With PointGoal Agents,” in *WACV*, 2024. URL: <https://3dlg-hcvc.github.io/mopa>.
- 4 S. Raychaudhuri, E. Cancelli, T. Campari, L. Ballan, M. Savva, and A. X. Chang, *LangNavBench: Evaluation of Natural Language Understanding in Semantic Navigation (under review)*, 2025. arXiv: [2507.07299](#). URL: <https://3dlg-hcvc.github.io/langmonmap/>.
- 5 S. Raychaudhuri and A. X. Chang, *Semantic Mapping in Indoor Embodied AI – A Survey on Advances, Challenges, and Future Directions (accepted to TMLR)*, 2025. arXiv: [2501.05750 \[cs.RO\]](#). URL: <https://arxiv.org/abs/2501.05750>.
- 6 M. Deitke, D. Batra, Y. Bisk, *et al.*, *Retrospectives on the embodied ai workshop*, 2022. arXiv: [2210.06849 \[cs.CV\]](#). URL: <https://arxiv.org/abs/2210.06849>.
- 7 F. Taioli, S. Yang, S. Raychaudhuri, M. Cristani, U. Jain, and A. X. Chang, *VISOR: Visual Spatial Object Reasoning for Language-driven Object Navigation*, 2026. arXiv: [2602.07555 \[cs.CV\]](#). URL: <https://arxiv.org/abs/2602.07555>.

Education

- 2019 – Fall 2025 **Ph.D., Simon Fraser University** in Computer Science. CGPA: 4.0 [[Thesis](#)]
- 2003 – 2007 **B.E., Indian Institute of Engineering Science and Technology, Shibpur (IEST)**
CGPA: 4.0

Employment History

Industry

- January – June 2024  **Research Intern.** Robotics and AI Institute, Cambridge, MA, USA.
Role: Conducted research on embodied AI and robotic perception/navigation, focusing on semantic understanding, instruction following, and zero-shot generalization in real-world environments.
Responsibilities:
- Designed and implemented hybrid autonomous solution for instruction-conditioned navigation that works seamlessly on both simulated environments and real-world robots.
 - Collaborated with cross-functional teams including research scientists and engineers.
 - Authored technical reports and contributed to research publications.
- 2011 – 2019  **Senior Software Engineer.** Ericsson India Global Services Pvt. Ltd., India.
Role: Led development of large-scale telecom software systems, delivering reliable, high-performance backend services and mentoring junior engineers.
- 2007 – 2011  **Software Engineer.** Tata Consultancy Services Pvt. Ltd., Kolkata, India.
Role: Developed enterprise software applications and supported client-facing systems in a large-scale IT services environment.

Teaching

- Spring 2026  **Graduate Teaching Assistant**, Natural Language Processing. SFU, Canada.
- Fall 2021  **Graduate Teaching Assistant**, Natural Language Processing. SFU, Canada.
- Spring 2021  **Graduate Teaching Assistant**, Natural Language Understanding. SFU, Canada.
- Fall 2020  **Graduate Teaching Assistant**, Natural Language Processing. SFU, Canada.

Miscellaneous Experience

Awards and Achievements

- 2022  **Outstanding TA Award** Computer Science, Simon Fraser University.

Peer Reviews

- Intl. Conferences.  TPAMI ([certificate](#)), ICRA, IROS, RA-L, CVPR, ICCV, ACL ARR, CRV, WACV, SIGGRAPH, SIGGRAPH Asia, IJCV, and more.

Others

- September 2025  **Attended** [PAISS Summer School @ INRIA](#).
- July 2025  **Volunteered** in [WiML Symposium @ ICML](#).
- 2024, 2023, 2022  **Co-hosted MultiON challenge.** [Embodied AI Workshop @ CVPR](#). ['24, '23, '22].
- 2023  **Co-organized** Visual Computing and Robotics (VCR) weekly seminars at SFU.
- 2022  **Organized** Reading Group at the 3dlg Lab in SFU.

References

Available on Request